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Honors Data Science

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Streets After Dark: Uncovering Crime Patterns In The Nations Capitol

1. Objective

This project looks at crime incident data from Washington, D.C. in 2024 to find patterns in crime types, locations, timing, and methods. It explores questions such as which crimes are most common, which wards have the most incidents, when crimes happen most often, what methods are used, and how crime volume changes during the year. By answering these questions, the analysis provides a clearer picture of public safety trends in the city.

2. Introduction

The Crime Incidents dataset comes from DC Open Data (opendata.dc.gov), managed by the District of Columbia government. It contains 29,303 reported crime incidents from 2024 and includes 25 variables, such as offense type, ward, time shift, method used, block location, and report date.

Studying crime patterns in Washington, D.C. is important for several reasons. DC is a busy, diverse city with big differences between its eight wards, and crime data shows where and when people are most at risk. This dataset is helpful because it connects location, time, and crime type, giving a deeper look at public safety. Analyzing this data can reveal insights that help residents, policymakers, and city planners.

3. Method

I used Python and Jupyter Notebook to do the analysis. The main libraries were pandas for handling data, numpy for calculations, and matplotlib and seaborn for making charts.

First, I loaded the dataset with `pd.read_csv()` and checked it using `.head()`, `.dtypes`, `.shape`, and `.isnull().sum()` to see its structure, column types, and missing data. The dataset has 29,303 rows and 25 columns, with key variables like OFFENSE, WARD, SHIFT, METHOD, and REPORT_DAT.

To clean the data, I removed duplicate rows with `.drop_duplicates()`. Then I dropped rows with missing values in key columns like `OFFENSE`, `WARD`, `SHIFT`, `METHOD`, and `REPORT_DAT` using `.dropna()`. After cleaning, 29,299 rows were left, showing that very little data was lost and the dataset was already high quality.

Next, I converted the `REPORT_DAT` column from a string to a datetime object using `pd.to_datetime()`. I also made three new columns: `MONTH` (number), `DAY_OF_WEEK` (name), and `HOUR` (hour). These changes made it possible to analyze the data by time.

4. Storytelling: Data Visualization and Interpretation

Figure 1 – Crime Incidents by Type

The first chart is a horizontal bar graph that shows how often each crime type appears in the data. THEFT/OTHER is the most common, with over 12,000 incidents, nearly twice as many as the next most common, THEFT F/AUTO (auto theft), which has about 6,000. MOTOR VEHICLE THEFT, ROBBERY, and ASSAULT W/DANGEROUS WEAPON follow. Violent crimes like HOMICIDE, SEX ABUSE, and ARSON are much less frequent and are at the bottom of the chart.

This pattern shows that most crime in DC is property crime, not violent crime. The high number of thefts suggests that things like parked cars and easy access to personal belongings are major reasons for crime in the city. While violent crimes are still serious, their lower numbers mean that most crime in DC is not violent.

Figure 2 – Crime Incidents by Ward

The second chart is a bar graph comparing the total number of crimes in each of DC's eight wards. Ward 2 has the most incidents, with Ward 1 close behind. Ward 3 has the fewest crimes by a large margin, and Wards 5, 6, and 7 are in the middle.

Ward 2 includes areas like Downtown DC, Dupont Circle, and Georgetown, which are busy commercial and tourist spots with large crowds. This likely explains why crime is higher there, since more people and activity increase the chances of theft. Ward 3 is mostly residential and wealthier, which may be why it has less crime. These differences show that police strategies should match each ward's unique needs.

Figure 3 – Crime Incidents by Time Shift

The third chart is a pie chart that shows crime incidents by time shift: Day, Evening, and Midnight. Evening makes up 40.5% of incidents, day is close behind at 39.3%, and midnight accounts for only 20.3%.

This result is a bit surprising because many people think crime is highest at night. But since most DC crime is property theft, it makes sense that crimes happen more during the day and evening when people are out. Fewer crimes at midnight do not mean nights are safe; it just shows that theft usually happens when people and their belongings are around. Figure 4 – Crime Incidents by Method Used

The fourth chart is a bar graph showing the method used in crimes: OTHERS, GUN, and KNIFE. Most crimes, over 25,000, are in the OTHERS category, meaning no weapon was used. Gun-related crimes are in the low thousands, and knife crimes are rare. This matches the earlier finding that most DC crime is property theft, which usually does not involve weapons. While gun crimes are less common, they are still a real safety concern, especially in cases like robbery and assault with a dangerous weapon. Knife crimes are the least common, so when violence does happen, it is more likely to involve guns than knives in DC. Figure 5 – Monthly Crime Trends

The fifth chart is a line graph showing crime numbers for each month in 2024. Crime is lower in January and February, rises in the spring, and peaks in July and August with over 2,800 incidents each month. It then drops through the fall and into December.

This seasonal trend is common in criminology research and is often linked to more outdoor activity during warmer months. When more people are outside, there are more chances for theft, assault, and other crimes. In winter, fewer people are outside, so there are fewer crimes. This suggests the city could add more police and resources in the summer to handle the increase in crime. The most important finding is that DC crime is mostly property-based: THEFT/OTHER is the main offense type, most crimes involve no weapon, and most incidents happen during the day and evening when people and their property are most accessible.

Looking at locations, Ward 2 has the most crime, likely because it has many businesses and tourists. Ward 3, which is more residential and wealthy, has the lowest crime rate. Over time, crime peaks in summer and drops in winter, a pattern seen in other crime studies too.

These findings have practical uses. Policymakers and law enforcement can use this data to better plan where and when to send patrols, like adding more officers in busy wards during evenings and summer, and focusing on theft prevention in crowded areas. Future studies could look at crime type by ward and time to find even more specific patterns. Overall, data science is a valuable tool for understanding and improving city safety.

6. References

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